

Haryana LEET / OCET 2026

Section C — Mechanical Stream Courses

20 Questions | 20 Marks | No Negative Marking

c-1 ELEMENTS OF MECHANICAL ENGINEERING (8 Q | 8 M)

■ Applied Mechanics — Friction

- Laws of dry friction: $F = \mu N$ | $\mu_s > \mu_k$
- Angle of friction $\phi = \tan^{-1}(\mu)$ | Angle of repose
- Applications: wedge, screw jack (torque = $W \cdot \tan(\phi + \lambda) \cdot r$)

■ Applied Mechanics — Centroid & Machines

- Centroid of: rectangle ($b/2, h/2$), triangle ($b/3, h/3$), circle (centre)
- Simple machines: $MA = \text{Load/Effort}$ | VR | $\eta = MA/VR$
- Types: wheel & axle $VR = R/r$ | Pulleys: $VR = \text{no. of rope segments}$
- Screw jack: $VR = 2\pi L/p$ | Projectile: $R = u^2 \sin 2\theta / g$, $H = u^2 \sin^2 \theta / 2g$

■ Strength of Materials

- Stress $\sigma = F/A$ | Strain $\epsilon = \delta L/L$ | $E = \sigma/\epsilon$ (Young's modulus)
- Poisson's ratio $\nu = \text{lateral strain} / \text{longitudinal strain}$
- Stress-strain diagram: elastic limit, yield point, UTS, fracture
- Thermal stress: $\sigma = E\alpha\Delta T$ (no expansion allowed)
- Relations: $E = 2G(1+\nu) = 3K(1-2\nu)$
- Resilience: ability to absorb energy elastically
- Principal stresses & principal planes — Mohr's circle
- SF & BM diagrams: simply supported beam (UDL, point load)
- Cantilever: max BM at fixed end | Columns: Euler's formula $P_{cr} = \pi^2 EI / L_e^2$

■ Thermodynamics

- Zeroth law: basis of temperature measurement (thermometry)
- 1st law: $Q = \Delta U + W$ | energy conservation
- 2nd law: Kelvin-Planck (no 100% heat to work) | Clausius statement
- Otto cycle (petrol engine): $\eta = 1 - 1/r^{(\gamma-1)}$ | r = compression ratio
- Diesel cycle: $\eta = 1 - (1/r^{(\gamma-1)}) \times (\rho^{(\gamma-1)} / (\gamma(p-1)))$
- Steam properties: dryness fraction $x = mg/(mf+mg)$
- Processes: isothermal (T const), adiabatic ($Q=0$), isobaric (P const), isochoric (V const)

■ Heat Transfer

- Conduction: $Q = kA(T_1 - T_2)/L$ (Fourier's law) | k = thermal conductivity
- Convection: $Q = hA(T_s - T_\infty)$ (Newton's law of cooling)
- Radiation: $Q = \epsilon\sigma AT^4$ (Stefan-Boltzmann) | $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$

■ Fluid Mechanics

- Density ρ , Specific gravity, Viscosity μ (dynamic), $\nu = \mu/\rho$ (kinematic)
- Newtonian fluid: $\tau = \mu(du/dy)$ | Non-Newtonian: $\tau \neq \mu(du/dy)$
- Bernoulli's theorem: $P/\rho g + v^2/2g + z = \text{constant}$ (per unit weight)
- Reynolds number $Re = \rho v D / \mu$ | $Re < 2100$ laminar | $Re > 4000$ turbulent
- Venturimeter: measures flow rate using pressure difference
- Pitot tube: measures flow velocity using stagnation pressure
- Darcy-Weisbach: $h_f = f L v^2 / 2gD$ | f = friction factor

c-2 ELEMENTS OF PRODUCTION ENGINEERING / MANUFACTURING (6 Q | 6 M)

■ Lathe Machine

- Operations: turning, facing, taper turning, threading, knurling, boring, parting
- Cutting speed $v = \pi DN/1000$ (m/min) | Feed f (mm/rev)
- Machining time $T = L/(f \times N)$ | D = dia (mm), N = RPM

■ Drilling & Milling

- Drilling operations: drilling, reaming, boring, counter-boring, counter-sinking, spot-facing
- Drilling time $T = L/(f \times N)$ | Sensitive, radial, pillar drill types
- Up-milling (conventional): tool against feed, poor surface finish
- Down-milling (climb): tool with feed, good surface finish
- Milling cutters: plain, face, end mill, form cutter, T-slot cutter

■ Shaper & Planer

- Shaper: reciprocating tool, stationary workpiece (small jobs)
- Planer: stationary tool, reciprocating table (large jobs)
- Quick return mechanism: forward stroke (cutting) slower than return

■ Measuring Instruments

- Vernier caliper: $LC = 1 \text{ MSD} - 1 \text{ VSD}$ (usually 0.02 mm)
- Micrometer: $LC = \text{pitch}/n = 0.5/50 = 0.01 \text{ mm}$
- Sine bar: sets precise angles using slip gauges
- Plug gauge (GO/NO-GO): checks hole size
- Snap gauge: checks shaft diameter
- Ring gauge: checks external diameter

■ Welding, Soldering, Brazing

- Arc welding: electric arc melts base metal (temp $\sim 3500^\circ\text{C}$)
- Gas welding: oxy-acetylene flame (neutral/oxidizing/carburizing)
- Resistance welding: spot, seam, projection welding
- Soldering: filler $< 450^\circ\text{C}$ (tin-lead) | Brazing: filler $> 450^\circ\text{C}$ (brass)

■ Material Science

- Mechanical properties: elasticity, plasticity, ductility, malleability, toughness, hardness
- Hardness testing: Brinell (BHN), Rockwell (HRC/HRB), Vickers (VHN)
- Heat treatment: Annealing (soften), Hardening (quench), Tempering (reduce brittleness)
- Normalizing: air cooling | Carburizing: adds carbon to surface

■ Engineering Drawing (Conceptual)

- 1st angle projection: object between observer & plane (used in India/Europe)
- 3rd angle projection: plane between observer & object (used in USA)
- Orthographic views: front, top, side views
- Isometric view: 3D representation on 2D at 30° axes

■ Foundry & Plastics

- Patterns: solid, split, match plate, cope & drag, sweep pattern
- Moulding sand properties: permeability, strength, refractoriness, cohesiveness
- Thermoplastic: can be re-melted (PVC, nylon, polyethylene)
- Thermosetting: cannot be re-melted (bakelite, epoxy, melamine)
- Plastic moulding: injection, compression, blow, transfer moulding

c-3 AUTOMOBILE ENGINEERING (6 Q | 6 M)

■ Power Flow in Automobile

- Engine $\rightarrow$ Clutch $\rightarrow$ Gearbox $\rightarrow$ Propeller shaft $\rightarrow$ Differential $\rightarrow$ Half shafts $\rightarrow$ Wheels

■ Clutch

- Function: smooth engagement/disengagement of engine & transmission
- Types: friction (single/multi-plate), fluid/hydraulic, electromagnetic
- Pressure plate, friction disc, release mechanism

■ Gearbox

- Function: vary torque & speed to suit road conditions
- Types: sliding mesh, constant mesh, synchromesh (manual)
- Automatic: epicyclic gearbox, torque converter | CVT: continuously variable

■ Brakes

- Drum brakes: leading-trailing shoe, duo-servo — used in rear wheels
- Disc brakes: caliper squeezes rotor — used in front wheels, better cooling
- Hydraulic braking: Pascal's law — master cylinder → wheel cylinders
- ABS (Anti-lock Braking System): prevents wheel lock during hard braking

■ Differential

- Function: allows left & right wheels to rotate at different speeds (cornering)
- Bevel gears: crown wheel, pinion, planet gears, sun gears

■ Cooling System

- Air-cooled: fins on cylinder, forced by fan — simple, lightweight
- Water-cooled: coolant (water+antifreeze) circulates through water jacket
- Radiator, thermostat, water pump, coolant reservoir

■ Lubrication System

- Functions: reduce friction, cooling, cleaning, sealing
- Wet sump: oil stored in sump pan (common)
- Dry sump: separate oil tank (racing cars)
- Methods: splash, pressure, combination

■ Wheels, Tyres & Steering

- Tyres: cross-ply (bias-ply), radial (most common), tubeless
- Tyre designation: 185/65 R15 (width/aspect ratio/radial/rim diameter)
- Steering systems: rack & pinion (cars), recirculating ball (trucks)
- Steering angles: camber, caster, toe-in/toe-out, king pin inclination

■ EV & Hybrid Vehicles

- BEV (Battery Electric): only electric motor, no IC engine
- HEV (Hybrid): IC engine + electric motor | Series hybrid vs Parallel hybrid
- PHEV: plug-in hybrid — can charge from external source
- Motors used: BLDC, PMSM | Battery: Li-ion, Li-Fe phosphate